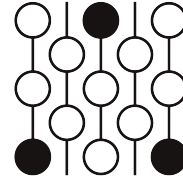


Review of Submission 8A

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Correctness. This friend gave the right construction, but didn't fully justify why it satisfies all the desired properties. Right after Proposition 1, it assumed a statement that is unknown to the community without any reference or justification. Even assuming that statement is true, the logical deduction has many missing parts.

Novelty. This friend adopted a different approach, which assumed a conjecture to be true.

2 Recommendation

I recommend that this submission should be rejected.

3 Submission with inline comments

A polar transform for the relative Dressian

o. Conventions and two local facts

Let E be the ground set, $r = \text{rk } M$, and μ_M the trivial rank r valuated matroid supported on the bases of M . I use the standard model of the relative Dressian: a rank k point is a projective valuated matroid

I don't know what 'projective valuated matroid' is.

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ν with $\nu \preceq \mu_M$, where $\preceq$ denotes valuated quotient. Brandt–Eur–Zhang [1] identify this order with inclusion of projective tropical linear spaces,

$$\text{Trop}(\theta) \subseteq \text{Trop}(\nu) \iff \theta \preceq \nu.$$

For valuated matroids p, q of ranks $a \leq b$, $p \preceq q$ is equivalent to the incidence-Plucker relations

$$\min_{i \in J \setminus I} \{p(I \cup i) + q(J \setminus i)\} \quad \text{attained at least twice} \quad (\text{IP})$$

for all $I \in \binom{E}{a-1}$ and $J \in \binom{E}{b+1}$. All- ∞ minima are regarded as attained at every displayed index.

Criterion 1 (nested incidence suffices). *Let p, q be valuated matroids of ranks $a < b$ whose ordinary supports are in quotient order. If (IP) holds whenever $I \subseteq J$, then (IP) holds for all I, J .*

Proof. For a rank a valuated matroid p and an $(a-1)$ -set I , write $C_p^*(I)_e = p(I \cup e)$, with value ∞ if $e \in I$. We use the standard cocircuit-generation theorem of Dress–Wenzel/Speyer [2, 5]:

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This is the wrong reference.

the tropical linear space of a valuated matroid is the closed tropical convex hull of its valuated cocircuits.

Fix arbitrary $I \in \binom{E}{a-1}$ and $J \in \binom{E}{b+1}$, and put $c_j = q(J \setminus j)$ for $j \in J$. If all c_j are ∞ there is nothing to prove. Otherwise J contains a q -basis. Since the ordinary support of p is a quotient of the ordinary support of q , every q -basis spans the support matroid of p (indeed, $a - \text{rk}_p(Q) \leq b - \text{rk}_q(Q) = 0$ for a q -basis Q); hence the restriction $p|_J$ has rank a .

For every $S \subseteq J$ with $|S| = a-1$, the nested hypothesis

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‘nest hypothesis’ should be (IP).

with upper set J says precisely that the cocircuit $C_{p|_J}^*(S)$ is tropically orthogonal to $c = (c_j)_{j \in J}$. The set of vectors orthogonal to a fixed c is a tropical hyperplane, hence is tropically convex; therefore it contains the whole tropical linear space of $p|_J$. It remains only to note that $C_p^*(I)|_J$ is a vector of $p|_J$:

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Do you mean ‘covector’?

for each valuated circuit of $p|_J$, indexed by a set $D \in \binom{J}{a+1}$, the orthogonality relation

$$\min_{e \in D} \{p(I \cup e) + p(D \setminus e)\}$$

is exactly a Plucker relation for p . Hence $C_p^*(I)|_J$ is orthogonal to c ,

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I don’t know how this conclusion is drawn. The equation above does not involve q . Hence, I don’t think the claim is proved.

which is the desired arbitrary incidence relation. $\square$

We also use Wang’s codimension-one theorem in the following precise form. In Wang’s terminology the “three-term incidence relations” consist of degenerate, parallel, and concurrent hyperplane relations; the displayed tropical minimum below is only the concurrent relation.

Proposition 1 (elementary quotient test). *Let η be a rank d valuated matroid and let $\theta : \binom{E}{d-1} \rightarrow \mathbb{R} \cup \{\infty\}$ be nonzero. Suppose that*

$$\theta(A) = \infty \quad \text{whenever } A \text{ is dependent in the support matroid of } \eta \quad (\text{D})$$

and, when $d \geq 2$, for every $S \in \binom{E}{d-2}$ and distinct $x, y, z \notin S$,

$$\min\{\theta(Sx) + \eta(Syz), \theta(Sy) + \eta(Sxz), \theta(Sz) + \eta(Sxy)\} \quad (\text{C})$$

is attained at least twice. Then θ is a valuated matroid and $\theta \preceq \eta$.

Proof. For $d = 1$ this is the unique rank-zero quotient. Let $d \geq 2$. By Wang's Theorem 4.3 [6], $\text{Dr}^1(\eta)$ is cut out by the degenerate, parallel, and concurrent three-term incidence relations. Condition (D) is exactly the degenerate part, and (C) contains the concurrent part. It remains only to see that the parallel part follows from (C). Write $A = Di$ and $B = Dj$ for independent $(d-1)$ -sets in the support matroid of η with the same closure, and choose k outside that closure, so Dik and Djk are bases. Applying (C) to D, i, j, k , the third upper coordinate $\eta(Dij)$ is ∞ , while $\eta(Dik)$ and $\eta(Djk)$ are finite. Hence

$$\theta(Di) + \eta(Djk) = \theta(Dj) + \eta(Dik),$$

which is Wang's parallel hyperplane relation. Thus all of Wang's defining relations hold, so θ represents a codimension-one tropical linear subspace of $\text{Trop}(\eta)$. $\square$

Finally, if $\nu \preceq \mu_M$ has rank k , we use its valuated Higgs completion

$$h_j(B) = \begin{cases} \min\{\nu(C) : B \subseteq C, |C| = k\}, & j \leq k, |B| = j, \\ \min\{\nu(C) : C \subseteq B, |C| = k\}, & j \geq k, B \in \mathcal{I}_j(M), \\ \infty, & j \geq k, B \notin \mathcal{I}_j(M). \end{cases} \quad (1)$$

Murota's truncation and relative elongation theorems [4, Theorems 2.1 and 4.1], equivalently the Dress–Terhalle well-layering construction [3],

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The construction is in [4], not [3]. Moreover, it has never been proved that it defines a completion for general matroids M .

give valuated matroids h_j and adjacent quotients $h_j \preceq h_{j+1}$. The top layer is projectively μ_M : if $B' = B - e + f$ are adjacent M -bases and $C \subseteq B$ minimizes (1), then either $e \notin C$ and $C \subseteq B'$, or the incidence relation for $\nu \preceq \mu_M$ with lower set $C - e$ and upper set $B \cup f$ gives a k -subset of B' of value at most $\nu(C)$; by symmetry the two top-layer values are equal.

1. Flat-lattice duality and flat constancy

Lemma 1. *If $\mathcal{L}(M)$ admits an order-reversing involution $X \mapsto X^\perp$, then $\mathcal{L}(M)$ is modular and $\text{rk}(X^\perp) = r - \text{rk}(X)$.*

Proof. The anti-automorphism reverses maximal chains, so it sends rank i to rank $r - i$. Applying it to the semimodular inequality for the geometric lattice $\mathcal{L}(M)$ gives the reverse inequality, hence equality. $\square$

Lemma 2 (flat constancy). *Let $\nu \preceq \mu_M$ have rank k . If B, B' are independent k -subsets of M with $\text{cl}(B) = \text{cl}(B')$, then $\nu(B) = \nu(B')$; and $\nu(A) = \infty$ whenever A is dependent in M .*

Proof. The support matroid of ν is an ordinary quotient of M , so its bases are independent in M . For $k > 0$ it suffices to compare $B = A \cup b$ and $B' = A \cup b'$ spanning the same rank k flat. Choose one complement C so that both AbC and $Ab'C$ are M -bases. In the incidence relation for $\nu \preceq \mu_M$ with lower set A and upper set $A \cup \{b, b'\} \cup C$, the only finite μ_M -terms are the b - and b' -terms. Thus $\nu(Ab) = \nu(Ab')$. $\square$

Write $\bar{\nu}(X)$ for the common value on independent k -sets spanning the rank k flat X , and put $\bar{\nu}(X) = \infty$ if no basis of ν spans X .

2. The polar transform

For a rank k quotient $\nu \preceq \mu_M$ define

$$\nu^{\perp M}(A) = \begin{cases} \bar{\nu}((\text{cl } A)^\perp), & A \text{ independent in } M, |A| = r - k, \\ \infty, & \text{otherwise.} \end{cases} \quad (2)$$

This is projectively well-defined, and applying it twice recovers the original flat coordinates.

Lemma 3 (ordinary polar rank). *Let N be an ordinary quotient of M of rank k , with rank function ρ_N . Then*

$$\rho^\perp(A) = \text{rk}_M(A) - k + \rho_N((\text{cl } A)^\perp) \quad (3)$$

is a matroid rank function. Its bases are precisely the finite support of (2). If N_p is a quotient of N_q , then $N_q^{\perp M}$ is a quotient of $N_p^{\perp M}$.

Proof. Since N is a quotient of M , ρ_N is constant on M -closures: if $e \in \text{cl}_M(A)$, then the N -rank increment from A to Ae is at most the corresponding M -increment, namely 0. First restrict (3) to flats. For flats $X \leq Y$,

$$\rho^\perp(Y) - \rho^\perp(X) = \text{rk}(Y) - \text{rk}(X) - (\rho_N(X^\perp) - \rho_N(Y^\perp))$$

is between 0 and $\text{rk}(Y) - \text{rk}(X)$ by the strong-map increment inequality, so $\rho^\perp$ is monotone with unit increments on flats. For flats X, Y , modularity of rk_M and the identities $(X \wedge Y)^\perp = X^\perp \vee Y^\perp$ and $(X \vee Y)^\perp = X^\perp \wedge Y^\perp$ reduce submodularity of $\rho^\perp$ to submodularity of ρ_N .

For arbitrary subsets, $\rho^\perp(A) = \rho^\perp(\text{cl}_M A)$. If $X = \text{cl } A$ and $Y = \text{cl } B$, then $\text{cl}(A \cap B) \leq X \wedge Y$ and $\text{cl}(A \cup B) = X \vee Y$; by monotonicity on flats, submodularity for X, Y implies submodularity for A, B . The unit-increment property also passes from flats to subsets. Thus (3) is a matroid rank function. An M -independent set A of size $r - k$ has full $\rho^\perp$ -rank exactly when $\rho_N((\text{cl } A)^\perp) = k$, and M -dependent sets cannot be bases; this is precisely the finite support of (2).

If N_p is a quotient of N_q , then on every interval the rank increment of N_p is at most that of N_q . The displayed formula for increments therefore reverses the inequality for the polar ranks, which is exactly the strong-map criterion saying that $N_q^{\perp M}$ is a quotient of $N_p^{\perp M}$. $\square$

Lemma 4 (elementary opposition). *Let $p \preceq q \preceq \mu_M$, with $\text{rk } p = d$ and $\text{rk } q = d + 1$. Then $q^{\perp M}$ and $p^{\perp M}$ satisfy all adjacent three-term relations of ranks $r - d - 1$ and $r - d$.*

Proof. Fix A of size $r - d - 2$ and distinct $x, y, z \notin A$. Put $U = \text{cl}(A)$, $W = \text{cl}(Axyz)$, and $\delta = \text{rk } W - \text{rk } U$. If A is dependent, or if $\delta \leq 1$, each term contains a dependent set of the required size and hence is ∞ .

If $\delta = 2$, let $G = W^\perp$ and $H = U^\perp$, of ranks d and $d + 2$. Whenever the term indexed by x is finite, Ayz spans W , so its p -factor is the common value $\bar{p}(G)$; its q -factor is $\bar{q}((\text{cl}(Ax))^\perp)$, and similarly for y, z . In the rank-two interval $[U, W]$, a loop over U makes all terms involving it infinite, and two parallel elements give equal corresponding q -coordinates. Thus only the case of three non-parallel points over U needs proof. Choose a basis S of G , representatives a_x, a_y, a_z for the three dual rank-one flats over G in $[G, H]$, and a complement C of H to a basis of M . In the incidence relation for $q \preceq \mu_M$ with lower set S and upper set $S \cup \{a_x, a_y, a_z\} \cup C$, the only finite μ_M -terms are those obtained by deleting one of a_x, a_y, a_z . Hence the three displayed q -coordinates have a non-unique minimum; adding the common $\bar{p}(G)$ proves the transformed relation.

If $\delta = 3$, then x, y, z form a Boolean frame over U . Let $G = W^\perp$ and

$$P_x = (\text{cl}(Ayz))^\perp, \quad P_y = (\text{cl}(Axz))^\perp, \quad P_z = (\text{cl}(Axy))^\perp.$$

These are the three atoms over G in the dual frame, and $(\text{cl}(Ax))^\perp = P_y \vee P_z$, with the analogous identities for y, z . Choose a basis S of G and representatives $a_e \in P_e \setminus G$. The adjacent incidence relation for $p \preceq q$ applied to the lower set S and to $S \cup \{a_x, a_y, a_z\}$ is exactly the transformed three-term relation, using flat constancy to identify the displayed coordinates. $\square$

Proposition 2. *The formula $\nu \mapsto \nu^{\perp M}$ defines an involution of the set underlying $\text{Dr}(M)$.*

Proof. Insert ν in the complete flag (1). We prove inductively that

$$h_r^{\perp M} \preceq h_{r-1}^{\perp M} \preceq \cdots \preceq h_0^{\perp M} = \mu_M.$$

After harmless scalar normalization, the right endpoint $h_0^{\perp M} = \mu_M$ is clear from (2). If $r \leq 1$, there is nothing more to check beyond the automatic rank-zero quotient. Now assume $r \geq 2$. Suppose $0 \leq j \leq r - 2$ and $h_j^{\perp M}$ has already been proved valuated. Since $h_j \preceq h_{j+1}$, Lemma 3 says that the ordinary support of $h_{j+1}^{\perp M}$ is an elementary quotient of the ordinary support of $h_j^{\perp M}$; in particular it satisfies the degenerate support condition (D) relative to $h_j^{\perp M}$. Lemma 4 gives the displayed concurrent relations (C) for the pair $h_{j+1}^{\perp M}, h_j^{\perp M}$. Proposition 1 then implies $h_{j+1}^{\perp M} \preceq h_j^{\perp M}$. This proves the induction down to $h_{r-1}^{\perp M}$; the last rank-zero quotient $h_r^{\perp M} \preceq h_{r-1}^{\perp M}$ is automatic. Hence $\nu^{\perp M}$, one layer of the flag, lies in $\text{Dr}(M)$. Involutivity follows from (2) and $(X^\perp)^\perp = X$. $\square$

3. Order reversal

Lemma 5 (circuit minima). *Let $w \preceq \mu_M$ have rank $k \geq 1$, let A be an independent $(k - 1)$ -set of M , and let C be a circuit of M/A with $|C| \geq 2$. Then $\min_{i \in C} w(A \cup i)$ is attained at least twice.*

Proof. Let $F = \text{cl}(A \cup C)$. For every $i \in C$, $A \cup (C - i)$ is independent and spans F . Choose one complement D of F to an M -basis. In the incidence relation for $w \preceq \mu_M$ with lower set A and upper set $A \cup C \cup D$, the finite μ_M -terms are exactly those indexed by $i \in C$. $\square$

Theorem 1. *If $p \preceq q \preceq \mu_M$, then*

$$q^{\perp_M} \preceq p^{\perp_M}.$$

Proof. Let $\text{rk } p = a$ and $\text{rk } q = b$. If $a = b$, the strong-map inequalities applied to $\emptyset \subseteq A$ and to $A \subseteq E$ force the two ordinary support rank functions to be equal. Then p and q are projectively equal: the equal-rank incidence relation on adjacent bases shows $p(B) - q(B)$ is constant on the common basis graph. If $a = 0$, then $p^{\perp_M} = \mu_M$, so the assertion is $q^{\perp_M} \preceq \mu_M$; if $b = r$, it is the vacuous rank-zero lower quotient. Hence assume $0 < a < b < r$.

By Proposition 2 the polars are valuated matroids, and by Lemma 3 their ordinary supports are in quotient order. By Criterion 1 it remains to prove nested incidence. Fix $A \subseteq T$ with $|A| = r - b - 1$, $|T| = r - a + 1$, set $B = T \setminus A$, and $m = |B| = b - a + 2$. If A is dependent all terms are ∞ . Otherwise put $U = \text{cl}(A)$, $W = \text{cl}(T)$, and $\delta = \text{rk } W - \text{rk } U$. A finite term forces $T - i$ to be independent, hence $\delta \geq m - 1$; since $\delta \leq m$, only two cases remain.

If $\delta = m$, then T is independent over U . Let $G = W^\perp$ and $H = U^\perp$, of ranks $a - 1$ and $b + 1$. The interval generated by B is Boolean; after applying $^\perp$, the flats $X_i = (\text{cl}(T - i))^\perp$ are the atoms over G , and $Y_i = (\text{cl}(A_i))^\perp$ is the join of all X_j with $j \neq i$. Choose a basis S of G and representatives $x_i \in X_i \setminus G$. The nested incidence relation for $p \preceq q$ with lower set S and upper set $S \cup \{x_i : i \in B\}$ is exactly the desired transformed relation.

If $\delta = m - 1$, then B has nullity one in M/A , hence a unique circuit C . A loop circuit gives only ∞ terms. Otherwise, for $i \notin C$ the set $T - i$ is dependent; for $i \in C$ it is independent and spans W , so the $p^{\perp_M}(T - i)$ factor is the common value $\bar{p}(W^\perp)$. Lemma 5, applied to $w = q^{\perp_M}$ and the circuit C of M/A , gives two minima among the remaining $q^{\perp_M}(A_i)$ terms. This proves nested incidence. $\square$

4. Agreement with flats

Let F be a flat of rank f , and let q_F be the trivial rank $r - f$ valuated matroid of $M/F \oplus U_{0,F}$. For an independent $(r - f)$ -set B with $Z = \text{cl}(B)$,

$$q_F(B) < \infty \iff Z \vee F = \hat{1}.$$

For an independent f -set A with $Y = \text{cl}(A)$, formula (2) gives

$$q_F^{\perp_M}(A) < \infty \iff Y^\perp \vee F = \hat{1}.$$

By modularity, since $\text{rk}(Y^\perp) = r - f$ and $\text{rk } F = f$, this is equivalent to $Y^\perp \wedge F = \hat{0}$. Applying the anti-automorphism gives $Y \vee F^\perp = \hat{1}$, the basis condition for $M/F^\perp \oplus U_{0,F^\perp}$. The finite values are zero on both sides. Thus the polar transform sends the flat F to $F^\perp$.

Combining Proposition 2, Theorem 1, and this flat computation, the map

$$\Phi([\nu]) = [\nu^{\perp_M}]$$

is an order-reversing involution of $\text{Dr}(M)$ extending the given order-reversing involution of $\mathcal{L}(M)$.

References

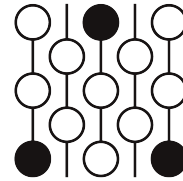
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- [2] A. Dress and W. Wenzel, *Valuated matroids*, Advances in Mathematics **93** (1992), 214–250.
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- [5] D. Speyer, *Tropical linear spaces*, SIAM Journal on Discrete Mathematics **22** (2008), 1527–1558.
- [6] J. Wang, *Lorentzian polynomials and the incidence geometry of tropical linear spaces*, arXiv:2412.12059v2 (2026).

Review of Submission 8A

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. This submission has many good ideas, some of which interestingly also appear in the human generated solution. However, the argument critically relies on the use of a valuated Higgs lift. It hallucinates references that do exist but which do not contain the statement it needs for a valuated Higgs lift, and this construction (to my knowledge) does not appear in the literature. However, with a proof of that statement, the strategy could be made to work.

Novelty. The proposed Higgs lift the submission falsely states is in the literature is actually a reasonable suggestion for the construction and is possibly correct in the case the larger matroid is trivially valuated.

2 Recommendation

This submission **requires major revisions**. The main fix it needs is to provide a proof that its hallucinated formula for the Higgs lift is valid (the middle matroid needs to be shown to be a valuated quotient). Assuming this (large) gap, the rest of the argument should go through. If it turns out that the Higgs lift hallucination cannot be made to work then this strategy doesn't have hope.

3 Submission with inline comments

A polar transform for the relative Dressian

o. Conventions and two local facts

Let E be the ground set, $r = \text{rk } M$, and μ_M the trivial rank r valuated matroid supported on the bases of M . I use the standard model of the relative Dressian: a rank k point is a projective valuated matroid ν with $\nu \preceq \mu_M$, where $\preceq$ denotes valuated quotient. Brandt–Eur–Zhang [1] identify this order with inclusion of projective tropical linear spaces,

$$\text{Trop}(\theta) \subseteq \text{Trop}(\nu) \iff \theta \preceq \nu.$$

For valuated matroids p, q of ranks $a \leq b$, $p \preceq q$ is equivalent to the incidence-Plucker relations

$$\min_{i \in J \setminus I} \{p(I \cup i) + q(J \setminus i)\} \quad \text{attained at least twice} \quad (\text{IP})$$

for all $I \in \binom{E}{a-1}$ and $J \in \binom{E}{b+1}$. All- ∞ minima are regarded as attained at every displayed index.

Criterion 1 (nested incidence suffices). *Let p, q be valuated matroids of ranks $a < b$ whose ordinary supports are in quotient order. If (IP) holds whenever $I \subseteq J$, then (IP) holds for all I, J .*

Proof. For a rank a valuated matroid p and an $(a-1)$ -set I , write $C_p^*(I)_e = p(I \cup e)$, with value ∞ if $e \in I$. We use the standard cocircuit-generation theorem of Dress–Wenzel/Speyer [2, 5]: the tropical linear space of a valuated matroid is the closed tropical convex hull of its valuated cocircuits.

Fix arbitrary $I \in \binom{E}{a-1}$ and $J \in \binom{E}{b+1}$, and put $c_j = q(J \setminus j)$ for $j \in J$. If all c_j are ∞ there is nothing to prove. Otherwise J contains a q -basis. Since the ordinary support of p is a quotient of the ordinary support of q , every q -basis spans the support matroid of p (indeed, $a - \text{rk}_p(Q) \leq b - \text{rk}_q(Q) = 0$ for a q -basis Q); hence the restriction $p|_J$ has rank a .

For every $S \subseteq J$ with $|S| = a-1$, the nested hypothesis with upper set J says precisely that the cocircuit $C_{p|_J}^*(S)$ is tropically orthogonal to $c = (c_j)_{j \in J}$. The set of vectors orthogonal to a fixed c is a tropical hyperplane, hence is tropically convex; therefore it contains the whole tropical linear space of $p|_J$. It remains only to note that $C_p^*(I)|_J$ is a vector of $p|_J$: for each valuated circuit of $p|_J$, indexed by a set $D \in \binom{J}{a+1}$, the orthogonality relation

$$\min_{e \in D} \{p(I \cup e) + p(D \setminus e)\}$$

is exactly a Plucker relation for p . Hence $C_p^*(I)|_J$ is orthogonal to c , which is the desired arbitrary incidence relation. $\square$

We also use Wang’s codimension-one theorem in the following precise form. In Wang’s terminology the “three-term incidence relations” consist of degenerate, parallel, and concurrent hyperplane relations; the displayed tropical minimum below is only the concurrent relation.

Proposition 1 (elementary quotient test). *Let η be a rank d valuated matroid and let $\theta : \binom{E}{d-1} \rightarrow \mathbb{R} \cup \{\infty\}$ be nonzero. Suppose that*

$$\theta(A) = \infty \quad \text{whenever } A \text{ is dependent in the support matroid of } \eta \quad (\text{D})$$

and, when $d \geq 2$, for every $S \in \binom{E}{d-2}$ and distinct $x, y, z \notin S$,

$$\min\{\theta(Sx) + \eta(Syz), \theta(Sy) + \eta(Sxz), \theta(Sz) + \eta(Sxy)\} \quad (\text{C})$$

is attained at least twice. Then θ is a valuated matroid and $\theta \preceq \eta$.

Proof. For $d = 1$ this is the unique rank-zero quotient. Let $d \geq 2$. By Wang’s Theorem 4.3 [6], $\text{Dr}^1(\eta)$ is cut out by the degenerate, parallel, and concurrent three-term incidence relations. Condition (D) is exactly the degenerate part, and (C) contains the concurrent part. It remains only to

see that the parallel part follows from (C). Write $A = Di$ and $B = Dj$ for independent $(d - 1)$ -sets in the support matroid of η with the same closure, and choose k outside that closure, so Dik and Djk are bases. Applying (C) to D, i, j, k , the third upper coordinate $\eta(Dij)$ is ∞ , while $\eta(Dik)$ and $\eta(Djk)$ are finite. Hence

$$\theta(Di) + \eta(Djk) = \theta(Dj) + \eta(Dik),$$

which is Wang's parallel hyperplane relation. Thus all of Wang's defining relations hold, so θ represents a codimension-one tropical linear subspace of $\text{Trop}(\eta)$. $\square$

Finally, if $\nu \preceq \mu_M$ has rank k , we use its valuated Higgs completion

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This is not a standard result in the literature and its citations do not contain a construction of this lift. It does not justify that the middle matroids are valuated quotients of μ_M . This is a load-bearing claim that is required to make this strategy work.

$$h_j(B) = \begin{cases} \min\{\nu(C) : B \subseteq C, |C| = k\}, & j \leq k, |B| = j, \\ \min\{\nu(C) : C \subseteq B, |C| = k\}, & j \geq k, B \in \mathcal{I}_j(M), \\ \infty, & j \geq k, B \notin \mathcal{I}_j(M). \end{cases} \quad (1)$$

Murota's truncation and relative elongation theorems [4, Theorems 2.1 and 4.1], equivalently the Dress–Terhalle well-layering construction [3], give valuated matroids h_j and adjacent quotients $h_j \preceq h_{j+1}$. The top layer is projectively μ_M : if $B' = B - e + f$ are adjacent M -bases and $C \subseteq B$ minimizes (1), then either $e \notin C$ and $C \subseteq B'$, or the incidence relation for $\nu \preceq \mu_M$ with lower set $C - e$ and upper set $B \cup f$ gives a k -subset of B' of value at most $\nu(C)$; by symmetry the two top-layer values are equal.

1. Flat-lattice duality and flat constancy

Reviewer 2

It is worth noting this submission does not go on and on about shorter lemmas that should not require a page to prove (unlike some other submissions).

Lemma 1. *If $\mathcal{L}(M)$ admits an order-reversing involution $X \mapsto X^\perp$, then $\mathcal{L}(M)$ is modular and $\text{rk}(X^\perp) = r - \text{rk}(X)$.*

Proof. The anti-automorphism reverses maximal chains, so it sends rank i to rank $r - i$. Applying it to the semimodular inequality for the geometric lattice $\mathcal{L}(M)$ gives the reverse inequality, hence equality. $\square$

Lemma 2 (flat constancy). *Let $\nu \preceq \mu_M$ have rank k . If B, B' are independent k -subsets of M with $\text{cl}(B) = \text{cl}(B')$, then $\nu(B) = \nu(B')$; and $\nu(A) = \infty$ whenever A is dependent in M .*

Proof. The support matroid of ν is an ordinary quotient of M , so its bases are independent in M . For $k > 0$ it suffices to compare $B = A \cup b$ and $B' = A \cup b'$ spanning the same rank k flat. Choose one complement C so that both AbC and $Ab'C$ are M -bases. In the incidence relation for $\nu \preceq \mu_M$ with lower set A and upper set $A \cup \{b, b'\} \cup C$, the only finite μ_M -terms are the b - and b' -terms. Thus $\nu(Ab) = \nu(Ab')$. $\square$

Write $\bar{\nu}(X)$ for the common value on independent k -sets spanning the rank k flat X , and put $\bar{\nu}(X) = \infty$ if no basis of ν spans X .

2. The polar transform

For a rank k quotient $\nu \preceq \mu_M$ define

$$\nu^{\perp_M}(A) = \begin{cases} \bar{\nu}((\text{cl } A)^\perp), & A \text{ independent in } M, |A| = r - k, \\ \infty, & \text{otherwise.} \end{cases} \quad (2)$$

This is projectively well-defined, and applying it twice recovers the original flat coordinates.

Lemma 3 (ordinary polar rank). *Let N be an ordinary quotient of M of rank k , with rank function ρ_N . Then*

$$\rho^\perp(A) = \text{rk}_M(A) - k + \rho_N((\text{cl } A)^\perp) \quad (3)$$

is a matroid rank function. Its bases are precisely the finite support of (2). If N_p is a quotient of N_q , then $N_q^{\perp_M}$ is a quotient of $N_p^{\perp_M}$.

Proof. Since N is a quotient of M , ρ_N is constant on M -closures: if $e \in \text{cl}_M(A)$, then the N -rank increment from A to Ae is at most the corresponding M -increment, namely 0. First restrict (3) to flats. For flats $X \leq Y$,

$$\rho^\perp(Y) - \rho^\perp(X) = \text{rk}(Y) - \text{rk}(X) - (\rho_N(X^\perp) - \rho_N(Y^\perp))$$

is between 0 and $\text{rk}(Y) - \text{rk}(X)$ by the strong-map increment inequality, so $\rho^\perp$ is monotone with unit increments on flats. For flats X, Y , modularity of rk_M and the identities $(X \wedge Y)^\perp = X^\perp \vee Y^\perp$ and $(X \vee Y)^\perp = X^\perp \wedge Y^\perp$ reduce submodularity of $\rho^\perp$ to submodularity of ρ_N .

For arbitrary subsets, $\rho^\perp(A) = \rho^\perp(\text{cl}_M A)$. If $X = \text{cl } A$ and $Y = \text{cl } B$, then $\text{cl}(A \cap B) \leq X \wedge Y$ and $\text{cl}(A \cup B) = X \vee Y$; by monotonicity on flats, submodularity for X, Y implies submodularity for A, B . The unit-increment property also passes from flats to subsets. Thus (3) is a matroid rank function. An M -independent set A of size $r - k$ has full $\rho^\perp$ -rank exactly when $\rho_N((\text{cl } A)^\perp) = k$, and M -dependent sets cannot be bases; this is precisely the finite support of (2).

If N_p is a quotient of N_q , then on every interval the rank increment of N_p is at most that of N_q . The displayed formula for increments therefore reverses the inequality for the polar ranks, which is exactly the strong-map criterion saying that $N_q^{\perp_M}$ is a quotient of $N_p^{\perp_M}$. $\square$

Lemma 4 (elementary opposition). *Let $p \preceq q \preceq \mu_M$, with $\text{rk } p = d$ and $\text{rk } q = d + 1$. Then $q^{\perp_M}$ and $p^{\perp_M}$ satisfy all adjacent three-term relations of ranks $r - d - 1$ and $r - d$.*

Proof. Fix A of size $r - d - 2$ and distinct $x, y, z \notin A$. Put $U = \text{cl}(A)$, $W = \text{cl}(Axyz)$, and $\delta = \text{rk } W - \text{rk } U$. If A is dependent, or if $\delta \leq 1$, each term contains a dependent set of the required size and hence is ∞ .

If $\delta = 2$, let $G = W^\perp$ and $H = U^\perp$, of ranks d and $d + 2$. Whenever the term indexed by x is finite, Ayz spans W , so its p -factor is the common value $\bar{p}(G)$; its q -factor is $\bar{q}((\text{cl}(Ax))^\perp)$, and similarly for y, z . In the rank-two interval $[U, W]$, a loop over U makes all terms involving it infinite, and two parallel elements give equal corresponding q -coordinates. Thus only the case of three non-parallel points over U needs proof. Choose a basis S of G , representatives a_x, a_y, a_z for the three dual rank-one flats over G in $[G, H]$, and a complement C of H to a basis of M . In the incidence relation for $q \preceq \mu_M$ with lower set S and upper set $S \cup \{a_x, a_y, a_z\} \cup C$, the only finite μ_M -terms are those obtained by deleting one of a_x, a_y, a_z . Hence the three displayed

Reviewer 2

Here the submission omits equations which it is using to complete the argument. For example it says the three “displayed” q -coordinates which it does not include. If the model uses some equations for internal reasoning it should include it if it is important for the argument, and it would improve the readability greatly.

q -coordinates have a non-unique minimum; adding the common $\bar{p}(G)$ proves the transformed relation.

If $\delta = 3$, then x, y, z form a Boolean frame over U . Let $G = W^\perp$ and

$$P_x = (\text{cl}(Ayz))^\perp, \quad P_y = (\text{cl}(Axz))^\perp, \quad P_z = (\text{cl}(Axy))^\perp.$$

These are the three atoms over G in the dual frame, and $(\text{cl}(Ax))^\perp = P_y \vee P_z$, with the analogous identities for y, z . Choose a basis S of G and representatives $a_e \in P_e \setminus G$. The adjacent incidence relation for $p \preceq q$ applied to the lower set S and to $S \cup \{a_x, a_y, a_z\}$ is exactly the transformed three-term relation, using flat constancy to identify the displayed coordinates. $\square$

Proposition 2. *The formula $\nu \mapsto \nu^{\perp M}$ defines an involution of the set underlying $\text{Dr}(M)$.*

Proof. Insert ν in the complete flag $(\mathbf{1})$. We prove inductively that

$$h_r^{\perp M} \preceq h_{r-1}^{\perp M} \preceq \cdots \preceq h_0^{\perp M} = \mu_M.$$

After harmless scalar normalization, the right endpoint $h_0^{\perp M} = \mu_M$ is clear from (2). If $r \leq 1$, there is nothing more to check beyond the automatic rank-zero quotient. Now assume $r \geq 2$. Suppose $0 \leq j \leq r - 2$ and $h_j^{\perp M}$ has already been proved valuated. Since $h_j \preceq h_{j+1}$, Lemma 3 says that the ordinary support of $h_{j+1}^{\perp M}$ is an elementary quotient of the ordinary support of $h_j^{\perp M}$; in particular it satisfies the degenerate support condition (D) relative to $h_j^{\perp M}$. Lemma 4 gives the displayed concurrent relations (C) for the pair $h_{j+1}^{\perp M}, h_j^{\perp M}$. Proposition 1 then implies $h_{j+1}^{\perp M} \preceq h_j^{\perp M}$. This proves the induction down to $h_{r-1}^{\perp M}$; the last rank-zero quotient $h_r^{\perp M} \preceq h_{r-1}^{\perp M}$ is automatic. Hence $\nu^{\perp M}$, one layer of the flag, lies in $\text{Dr}(M)$. Involutivity follows from (2) and $(X^\perp)^\perp = X$. $\square$

3. Order reversal

Lemma 5 (circuit minima). *Let $w \preceq \mu_M$ have rank $k \geq 1$, let A be an independent $(k - 1)$ -set of M , and let C be a circuit of M/A with $|C| \geq 2$. Then $\min_{i \in C} w(A \cup i)$ is attained at least twice.*

Proof. Let $F = \text{cl}(A \cup C)$. For every $i \in C$, $A \cup (C - i)$ is independent and spans F . Choose one complement D of F to an M -basis. In the incidence relation for $w \preceq \mu_M$ with lower set A and upper set $A \cup C \cup D$, the finite μ_M -terms are exactly those indexed by $i \in C$. $\square$

Reviewer 2

The following proof and the (elementary opposition) proof could possibly be unified so that it isn't as repetitive, but not a correctness problem.

Theorem 1. *If $p \preceq q \preceq \mu_M$, then*

$$q^{\perp_M} \preceq p^{\perp_M}.$$

Proof. Let $\text{rk } p = a$ and $\text{rk } q = b$. If $a = b$, the strong-map inequalities applied to $\emptyset \subseteq A$ and to $A \subseteq E$ force the two ordinary support rank functions to be equal. Then p and q are projectively equal: the equal-rank incidence relation on adjacent bases shows $p(B) - q(B)$ is constant on the common basis graph. If $a = 0$, then $p^{\perp_M} = \mu_M$, so the assertion is $q^{\perp_M} \preceq \mu_M$; if $b = r$, it is the vacuous rank-zero lower quotient. Hence assume $0 < a < b < r$.

By Proposition 2 the polars are valuated matroids, and by Lemma 3 their ordinary supports are in quotient order. By Criterion 1 it remains to prove nested incidence. Fix $A \subseteq T$ with $|A| = r - b - 1$, $|T| = r - a + 1$, set $B = T \setminus A$, and $m = |B| = b - a + 2$. If A is dependent all terms are ∞ . Otherwise put $U = \text{cl}(A)$, $W = \text{cl}(T)$, and $\delta = \text{rk } W - \text{rk } U$. A finite term forces $T - i$ to be independent, hence $\delta \geq m - 1$; since $\delta \leq m$, only two cases remain.

If $\delta = m$, then T is independent over U . Let $G = W^\perp$ and $H = U^\perp$, of ranks $a - 1$ and $b + 1$. The interval generated by B is Boolean; after applying $^\perp$, the flats $X_i = (\text{cl}(T - i))^\perp$ are the atoms over G , and $Y_i = (\text{cl}(Ai))^\perp$ is the join of all X_j with $j \neq i$. Choose a basis S of G and representatives $x_i \in X_i \setminus G$. The nested incidence relation for $p \preceq q$ with lower set S and upper set $S \cup \{x_i : i \in B\}$ is exactly the desired transformed relation.

If $\delta = m - 1$, then B has nullity one in M/A , hence a unique circuit C . A loop circuit gives only ∞ terms. Otherwise, for $i \notin C$ the set $T - i$ is dependent; for $i \in C$ it is independent and spans W , so the $p^{\perp_M}(T - i)$ factor is the common value $\bar{p}(W^\perp)$. Lemma 5, applied to $w = q^{\perp_M}$ and the circuit C of M/A , gives two minima among the remaining $q^{\perp_M}(Ai)$ terms. This proves nested incidence. $\square$

4. Agreement with flats

Let F be a flat of rank f , and let q_F be the trivial rank $r - f$ valuated matroid of $M/F \oplus U_{0,F}$. For an independent $(r - f)$ -set B with $Z = \text{cl}(B)$,

$$q_F(B) < \infty \iff Z \vee F = \hat{1}.$$

For an independent f -set A with $Y = \text{cl}(A)$, formula (2) gives

$$q_F^{\perp M}(A) < \infty \iff Y^\perp \vee F = \hat{1}.$$

By modularity, since $\text{rk}(Y^\perp) = r - f$ and $\text{rk } F = f$, this is equivalent to $Y^\perp \wedge F = \hat{0}$. Applying the anti-automorphism gives $Y \vee F^\perp = \hat{1}$, the basis condition for $M/F^\perp \oplus U_{0,F^\perp}$. The finite values are zero on both sides. Thus the polar transform sends the flat F to $F^\perp$.

Combining Proposition 2, Theorem 1, and this flat computation, the map

$$\Phi([\nu]) = [\nu^{\perp M}]$$

is an order-reversing involution of $\text{Dr}(M)$ extending the given order-reversing involution of $\mathcal{L}(M)$.

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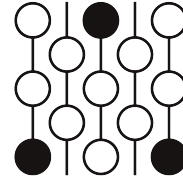
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- [2] A. Dress and W. Wenzel, *Valuated matroids*, Advances in Mathematics **93** (1992), 214–250.
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Review of Submission 8A

First Proof Second Batch

Reviewer 3

June 4–8, 2026



1 Review

Correctness. The proof is essentially correct with one major gap. The equation labeled (1) is falsely stated as following from the Work of Murota and Dress–Terhalle. The latter citation is a complete red herring. The former only deals with the well-known case where M is the Boolean matroid. The more general statement here does not seem to be in the literature, although it may be known to experts. It is related to a well-known open problem in the field. Its use must be justified by a proof that the construction actually results in valuated matroids and that these satisfy the desired quotient relations. Both are true but left completely unjustified.

Novelty. There are a number of interesting ideas in this proof. Criterion 1 is well-known to experts, but its proof here is remarkably efficient. The proof is a very delicate tapestry. The proof is geared towards proving the full result and would not be able to prove a partial result if one step were to fail. This contrasts with the human proof. Otherwise the ideas run parallel to the human proof. The necessary components are present, except for the proof of (1) mentioned above.

2 Recommendation

This submission **requires major revision** insofar as the proof of the assertion in (1) of a relative Higgs lift for a pair of valuated matroid is asserted when one of the two valuations is trivial. This surprising fact requires a proof. The false citation will not do.

3 Submission with inline comments

A polar transform for the relative Dressian

o. Conventions and two local facts

Let E be the ground set, $r = \text{rk } M$, and μ_M the trivial rank r valuated matroid supported on the bases of M . I use the standard model of the relative Dressian: a rank k point is a projective valuated

matroid ν with $\nu \preceq \mu_M$, where $\preceq$ denotes valuated quotient. Brandt–Eur–Zhang [1] identify this order with inclusion of projective tropical linear spaces,

$$\text{Trop}(\theta) \subseteq \text{Trop}(\nu) \iff \theta \preceq \nu.$$

For valuated matroids p, q of ranks $a \leq b$, $p \preceq q$ is equivalent to the incidence-Plucker relations

$$\min_{i \in J \setminus I} \{p(I \cup i) + q(J \setminus i)\} \text{ attained at least twice} \quad (\text{IP})$$

for all $I \in \binom{E}{a-1}$ and $J \in \binom{E}{b+1}$. All- ∞ minima are regarded as attained at every displayed index.

Criterion 1 (nested incidence suffices). *Let p, q be valuated matroids of ranks $a < b$ whose ordinary supports are in quotient order. If (IP) holds whenever $I \subseteq J$, then (IP) holds for all I, J .*

Proof. For a rank a valuated matroid p and an $(a-1)$ -set I , write $C_p^*(I)_e = p(I \cup e)$, with value ∞ if $e \in I$. We use the standard cocircuit-generation theorem of Dress–Wenzel/Speyer [2, 5]: the tropical linear space of a valuated matroid is the closed tropical convex hull of its valuated cocircuits.

Fix arbitrary $I \in \binom{E}{a-1}$ and $J \in \binom{E}{b+1}$, and put $c_j = q(J \setminus j)$ for $j \in J$. If all c_j are ∞ there is nothing to prove. Otherwise J contains a q -basis. Since the ordinary support of p is a quotient of the ordinary support of q , every q -basis spans the support matroid of p (indeed, $a - \text{rk}_p(Q) \leq b - \text{rk}_q(Q) = 0$ for a q -basis Q); hence the restriction $p|_J$ has rank a .

For every $S \subseteq J$ with $|S| = a-1$, the nested hypothesis with upper set J says precisely that the cocircuit $C_{p|_J}^*(S)$ is tropically orthogonal to $c = (c_j)_{j \in J}$. The set of vectors orthogonal to a fixed c is a tropical hyperplane, hence is tropically convex; therefore it contains the whole tropical linear space of $p|_J$. It remains only to note that $C_p^*(I)|_J$ is a vector of $p|_J$: for each valuated circuit of $p|_J$, indexed by a set $D \in \binom{J}{a+1}$, the orthogonality relation

$$\min_{e \in D} \{p(I \cup e) + p(D \setminus e)\}$$

is exactly a Plucker relation for p . Hence $C_p^*(I)|_J$ is orthogonal to c , which is the desired arbitrary incidence relation. $\square$

Reviewer 3

The above statement is known to experts but does not seem to appear in the literature. The proof provided is quite beautiful. However, the writer does not explain the argument very well in the last paragraph. The emphasis in the proof is all wrong, leading to a rather confusing read. The structure of the second paragraph should be:

- State that we will show that every covector of p is tropically orthogonal to every circuit of q .
- Fix a circuit C and extend to a spanning set J of size $\text{rk } q + 1$.
- Restrict to J . Reduce to using cocircuits of $p|_J$.
- Explain that this case is exactly handled by the hypothesis.

We also use Wang’s codimension-one theorem in the following precise form. In Wang’s terminology the “three-term incidence relations” consist of degenerate, parallel, and concurrent hyperplane relations; the displayed tropical minimum below is only the concurrent relation.

Proposition 1 (elementary quotient test). *Let η be a rank d valuated matroid and let $\theta : \binom{E}{d-1} \rightarrow \mathbb{R} \cup \{\infty\}$ be nonzero. Suppose that*

$$\theta(A) = \infty \quad \text{whenever } A \text{ is dependent in the support matroid of } \eta \quad (\text{D})$$

and, when $d \geq 2$, for every $S \in \binom{E}{d-2}$ and distinct $x, y, z \notin S$,

$$\min\{\theta(Sx) + \eta(Syz), \theta(Sy) + \eta(Sxz), \theta(Sz) + \eta(Sxy)\} \quad (\text{C})$$

is attained at least twice. Then θ is a valuated matroid and $\theta \preceq \eta$.

Reviewer 3

The reduction from Wang's statement to this one is essentially trivial. The proof could be omitted, or left to a single explanatory sentence instead of a formal proof environment.

Proof. For $d = 1$ this is the unique rank-zero quotient. Let $d \geq 2$. By Wang's Theorem 4.3 [6], $\text{Dr}^1(\eta)$ is cut out by the degenerate, parallel, and concurrent three-term incidence relations. Condition (D) is exactly the degenerate part, and (C) contains the concurrent part. It remains only to see that the parallel part follows from (C). Write $A = Di$ and $B = Dj$ for independent $(d-1)$ -sets in the support matroid of η with the same closure, and choose k outside that closure, so Dik and Djk are bases. Applying (C) to D, i, j, k , the third upper coordinate $\eta(Dij)$ is ∞ , while $\eta(Dik)$ and $\eta(Djk)$ are finite. Hence

$$\theta(Di) + \eta(Djk) = \theta(Dj) + \eta(Dik),$$

which is Wang's parallel hyperplane relation. Thus all of Wang's defining relations hold, so θ represents a codimension-one tropical linear subspace of $\text{Trop}(\eta)$. $\square$

Finally, if $\nu \preceq \mu_M$ has rank k , we use its valuated Higgs completion

$$h_j(B) = \begin{cases} \min\{\nu(C) : B \subseteq C, |C| = k\}, & j \leq k, |B| = j, \\ \min\{\nu(C) : C \subseteq B, |C| = k\}, & j \geq k, B \in \mathcal{I}_j(M), \\ \infty, & j \geq k, B \notin \mathcal{I}_j(M). \end{cases} \quad (1)$$

Murota's truncation and relative elongation theorems [4, Theorems 2.1 and 4.1], equivalently the Dress–Terhalle well-layering construction [3], give valuated matroids h_j and adjacent quotients $h_j \preceq h_{j+1}$. The top layer is projectively μ_M : if $B' = B - e + f$ are adjacent M -bases and $C \subseteq B$ minimizes (1), then either $e \notin C$ and $C \subseteq B'$, or the incidence relation for $\nu \preceq \mu_M$ with lower set $C - e$ and upper set $B \cup f$ gives a k -subset of B' of value at most $\nu(C)$; by symmetry the two top-layer values are equal.

Reviewer 3

Equation (1) is perhaps the most remarkable aspect of this submission. The Higgs lift is well-known in the case of ordinary matroids. The construction given here is also well-known when M is a Boolean matroid. However, the given construction, which generalizes both simultaneously does not seem to be

in the literature. The citations provided are bogus. There is also no justification provided as to why the functions h_j are matroids, nor why the quotient relation $h_{j+1} \twoheadrightarrow h_j$ holds. These are non-trivial to prove. However, they could be helped by repeated applications of Criterion 1.

1. Flat-lattice duality and flat constancy

Lemma 1. *If $\mathcal{L}(M)$ admits an order-reversing involution $X \mapsto X^\perp$, then $\mathcal{L}(M)$ is modular and $\text{rk}(X^\perp) = r - \text{rk}(X)$.*

Proof. The anti-automorphism reverses maximal chains, so it sends rank i to rank $r - i$. Applying it to the semimodular inequality for the geometric lattice $\mathcal{L}(M)$ gives the reverse inequality, hence equality. $\square$

Lemma 2 (flat constancy). *Let $\nu \preceq \mu_M$ have rank k . If B, B' are independent k -subsets of M with $\text{cl}(B) = \text{cl}(B')$, then $\nu(B) = \nu(B')$; and $\nu(A) = \infty$ whenever A is dependent in M .*

Proof. The support matroid of ν is an ordinary quotient of M , so its bases are independent in M . For $k > 0$ it suffices to compare $B = A \cup b$ and $B' = A \cup b'$ spanning the same rank k flat. Choose one complement C so that both AbC and $Ab'C$ are M -bases. In the incidence relation for $\nu \preceq \mu_M$ with lower set A and upper set $A \cup \{b, b'\} \cup C$, the only finite μ_M -terms are the b - and b' -terms. Thus $\nu(Ab) = \nu(Ab')$. $\square$

Reviewer 3

Lemmas 1 and 2 have the correct degree of concision. Still nicely explained.

Write $\bar{\nu}(X)$ for the common value on independent k -sets spanning the rank k flat X , and put $\bar{\nu}(X) = \infty$ if no basis of ν spans X .

2. The polar transform

For a rank k quotient $\nu \preceq \mu_M$ define

$$\nu^{\perp_M}(A) = \begin{cases} \bar{\nu}((\text{cl } A)^\perp), & A \text{ independent in } M, |A| = r - k, \\ \infty, & \text{otherwise.} \end{cases} \quad (2)$$

This is projectively well-defined, and applying it twice recovers the original flat coordinates.

Reviewer 3

This is the correct definition. It should be made clearer that cl means the closure in M .

Lemma 3 (ordinary polar rank). *Let N be an ordinary quotient of M of rank k , with rank function ρ_N . Then*

$$\rho^\perp(A) = \text{rk}_M(A) - k + \rho_N((\text{cl } A)^\perp) \quad (3)$$

is a matroid rank function. Its bases are precisely the finite support of (2). If N_p is a quotient of N_q , then $N_q^{\perp_M}$ is a quotient of $N_p^{\perp_M}$.

Proof. Since N is a quotient of M , ρ_N is constant on M -closures: if $e \in \text{cl}_M(A)$, then the N -rank increment from A to Ae is at most the corresponding M -increment, namely 0. First restrict (3) to flats. For flats $X \leq Y$,

$$\rho^\perp(Y) - \rho^\perp(X) = \text{rk}(Y) - \text{rk}(X) - (\rho_N(X^\perp) - \rho_N(Y^\perp))$$

is between 0 and $\text{rk}(Y) - \text{rk}(X)$ by the strong-map increment inequality, so $\rho^\perp$ is monotone with unit increments on flats. For flats X, Y , modularity of rk_M and the identities $(X \wedge Y)^\perp = X^\perp \vee Y^\perp$ and $(X \vee Y)^\perp = X^\perp \wedge Y^\perp$ reduce submodularity of $\rho^\perp$ to submodularity of ρ_N .

For arbitrary subsets, $\rho^\perp(A) = \rho^\perp(\text{cl}_M A)$. If $X = \text{cl } A$ and $Y = \text{cl } B$, then $\text{cl}(A \cap B) \leq X \wedge Y$ and $\text{cl}(A \cup B) = X \vee Y$; by monotonicity on flats, submodularity for X, Y implies submodularity for A, B . The unit-increment property also passes from flats to subsets. Thus (3) is a matroid rank function. An M -independent set A of size $r - k$ has full $\rho^\perp$ -rank exactly when $\rho_N((\text{cl } A)^\perp) = k$, and M -dependent sets cannot be bases; this is precisely the finite support of (2).

If N_p is a quotient of N_q , then on every interval the rank increment of N_p is at most that of N_q . The displayed formula for increments therefore reverses the inequality for the polar ranks, which is exactly the strong-map criterion saying that $N_q^{\perp M}$ is a quotient of $N_p^{\perp M}$. $\square$

Lemma 4 (elementary opposition). *Let $p \preceq q \preceq \mu_M$, with $\text{rk } p = d$ and $\text{rk } q = d + 1$. Then $q^{\perp M}$ and $p^{\perp M}$ satisfy all adjacent three-term relations of ranks $r - d - 1$ and $r - d$.*

Proof. Fix A of size $r - d - 2$ and distinct $x, y, z \notin A$. Put $U = \text{cl}(A)$, $W = \text{cl}(Axyz)$, and $\delta = \text{rk } W - \text{rk } U$. If A is dependent, or if $\delta \leq 1$, each term contains a dependent set of the required size and hence is ∞ .

If $\delta = 2$, let $G = W^\perp$ and $H = U^\perp$, of ranks d and $d + 2$. Whenever the term indexed by x is finite, Ayz spans W , so its p -factor is the common value $\bar{p}(G)$; its q -factor is $\bar{q}((\text{cl}(Ax))^\perp)$, and similarly for y, z . In the rank-two interval $[U, W]$, a loop over U makes all terms involving it infinite, and two parallel elements give equal corresponding q -coordinates. Thus only the case of three non-parallel points over U needs proof. Choose a basis S of G , representatives a_x, a_y, a_z for the three dual rank-one flats over G in $[G, H]$, and a complement C of H to a basis of M . In the incidence relation for $q \preceq \mu_M$ with lower set S and upper set $S \cup \{a_x, a_y, a_z\} \cup C$, the only finite μ_M -terms are those obtained by deleting one of a_x, a_y, a_z . Hence the three displayed q -coordinates have a non-unique minimum; adding the common $\bar{p}(G)$ proves the transformed relation.

If $\delta = 3$, then x, y, z form a Boolean frame over U . Let $G = W^\perp$ and

$$P_x = (\text{cl}(Ayz))^\perp, \quad P_y = (\text{cl}(Axz))^\perp, \quad P_z = (\text{cl}(Axy))^\perp.$$

These are the three atoms over G in the dual frame, and $(\text{cl}(Ax))^\perp = P_y \vee P_z$, with the analogous identities for y, z . Choose a basis S of G and representatives $a_e \in P_e \setminus G$. The adjacent incidence relation for $p \preceq q$ applied to the lower set S and to $S \cup \{a_x, a_y, a_z\}$ is exactly the transformed three-term relation, using flat constancy to identify the displayed coordinates. $\square$

In the above lemma, once gets the sense that there is a picture being drawn that is not shown to the reader. Either a physical depiction and/or writing down the actual equations would make the argument more palatable. This proof is correct but the reasoning is obscured by the failure to highlight the facts that make the argument work.

Proposition 2. *The formula $\nu \mapsto \nu^{\perp M}$ defines an involution of the set underlying $\text{Dr}(M)$.*

Proof. Insert ν in the complete flag (1). We prove inductively that

$$h_r^{\perp M} \preceq h_{r-1}^{\perp M} \preceq \cdots \preceq h_0^{\perp M} = \mu_M.$$

After harmless scalar normalization, the right endpoint $h_0^{\perp M} = \mu_M$ is clear from (2). If $r \leq 1$, there is nothing more to check beyond the automatic rank-zero quotient. Now assume $r \geq 2$. Suppose $0 \leq j \leq r-2$ and $h_j^{\perp M}$ has already been proved valuated. Since $h_j \preceq h_{j+1}$, Lemma 3 says that the ordinary support of $h_{j+1}^{\perp M}$ is an elementary quotient of the ordinary support of $h_j^{\perp M}$; in particular it satisfies the degenerate support condition (D) relative to $h_j^{\perp M}$. Lemma 4 gives the displayed concurrent relations (C) for the pair $h_{j+1}^{\perp M}, h_j^{\perp M}$. Proposition 1 then implies $h_{j+1}^{\perp M} \preceq h_j^{\perp M}$. This proves the induction down to $h_{r-1}^{\perp M}$; the last rank-zero quotient $h_r^{\perp M} \preceq h_{r-1}^{\perp M}$ is automatic. Hence $\nu^{\perp M}$, one layer of the flag, lies in $\text{Dr}(M)$. Involutivity follows from (2) and $(X^\perp)^\perp = X$. $\square$

This is rather delicately orchestrated argument. It relies crucially on the unproven assertion that the flag (1) exists.

3. Order reversal

Lemma 5 (circuit minima). *Let $w \preceq \mu_M$ have rank $k \geq 1$, let A be an independent $(k-1)$ -set of M , and let C be a circuit of M/A with $|C| \geq 2$. Then $\min_{i \in C} w(A \cup i)$ is attained at least twice.*

Proof. Let $F = \text{cl}(A \cup C)$. For every $i \in C$, $A \cup (C - i)$ is independent and spans F . Choose one complement D of F to an M -basis. In the incidence relation for $w \preceq \mu_M$ with lower set A and upper set $A \cup C \cup D$, the finite μ_M -terms are exactly those indexed by $i \in C$. $\square$

The above lemma is completely trivial and standard. It should be stated, not proven.

Theorem 1. *If $p \preceq q \preceq \mu_M$, then*

$$q^{\perp M} \preceq p^{\perp M}.$$

Proof. Let $\text{rk } p = a$ and $\text{rk } q = b$. If $a = b$, the strong-map inequalities applied to $\emptyset \subseteq A$ and to $A \subseteq E$ force the two ordinary support rank functions to be equal. Then p and q are projectively equal: the equal-rank incidence relation on adjacent bases shows $p(B) - q(B)$ is constant on the common basis graph. If $a = 0$, then $p^{\perp M} = \mu_M$, so the assertion is $q^{\perp M} \preceq \mu_M$; if $b = r$, it is the vacuous rank-zero lower quotient. Hence assume $0 < a < b < r$.

By Proposition 2 the polars are valuated matroids, and by Lemma 3 their ordinary supports are in quotient order. By Criterion 1 it remains to prove nested incidence. Fix $A \subseteq T$ with $|A| = r - b - 1$, $|T| = r - a + 1$, set $B = T \setminus A$, and $m = |B| = b - a + 2$. If A is dependent all terms are ∞ . Otherwise put $U = \text{cl}(A)$, $W = \text{cl}(T)$, and $\delta = \text{rk } W - \text{rk } U$. A finite term forces $T - i$ to be independent, hence $\delta \geq m - 1$; since $\delta \leq m$, only two cases remain.

If $\delta = m$, then T is independent over U . Let $G = W^\perp$ and $H = U^\perp$, of ranks $a - 1$ and $b + 1$. The interval generated by B is Boolean; after applying $^\perp$, the flats $X_i = (\text{cl}(T - i))^\perp$ are the atoms over G , and $Y_i = (\text{cl}(Ai))^\perp$ is the join of all X_j with $j \neq i$. Choose a basis S of G and representatives $x_i \in X_i \setminus G$. The nested incidence relation for $p \preceq q$ with lower set S and upper set $S \cup \{x_i : i \in B\}$ is exactly the desired transformed relation.

If $\delta = m - 1$, then B has nullity one in M/A , hence a unique circuit C . A loop circuit gives only ∞ terms. Otherwise, for $i \notin C$ the set $T - i$ is dependent; for $i \in C$ it is independent and spans W , so the $p^{\perp M}(T - i)$ factor is the common value $\bar{p}(W^\perp)$. Lemma 5, applied to $w = q^{\perp M}$ and the circuit C of M/A , gives two minima among the remaining $q^{\perp M}(Ai)$ terms. This proves nested incidence. $\square$

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This proof is essentially the same as the one given in Lemma 4. They should really be combined. There are subtle differences: here the rank gap is arbitrary and there it is not assumed that p, q are valuated matroids. However, this is a minor difference that can be handled by an appropriate formulation. The handling of the index i in this proof is less than ideal.

4. Agreement with flats

Let F be a flat of rank f , and let q_F be the trivial rank $r - f$ valuated matroid of $M/F \oplus U_{0,F}$. For an independent $(r - f)$ -set B with $Z = \text{cl}(B)$,

$$q_F(B) < \infty \iff Z \vee F = \hat{1}.$$

For an independent f -set A with $Y = \text{cl}(A)$, formula (2) gives

$$q_F^{\perp M}(A) < \infty \iff Y^\perp \vee F = \hat{1}.$$

By modularity, since $\text{rk}(Y^\perp) = r - f$ and $\text{rk } F = f$, this is equivalent to $Y^\perp \wedge F = \hat{0}$. Applying the anti-automorphism gives $Y \vee F^\perp = \hat{1}$, the basis condition for $M/F^\perp \oplus U_{0,F^\perp}$. The finite values are zero on both sides. Thus the polar transform sends the flat F to $F^\perp$.

Combining Proposition 2, Theorem 1, and this flat computation, the map

$$\Phi([\nu]) = [\nu^{\perp M}]$$

is an order-reversing involution of $\text{Dr}(M)$ extending the given order-reversing involution of $\mathcal{L}(M)$.

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